

1. Project Story / Case Study

Airbnb NYC 2024 — Exploratory Data Analysis

The Story Behind the Project

New York City is one of the world's busiest travel destinations, and Airbnb has a wide variety of listings across different neighbourhoods, room types, price ranges, and availability levels.

The goal of this project was to turn raw Airbnb listing data into meaningful insights by answering questions such as:

- Which NYC neighbourhood groups have higher average prices?
- How does room type influence listing prices?
- How is Airbnb pricing distributed?
- How does availability vary across listings?
- Does the number of reviews show a relationship with price?
- How are Airbnb listings geographically distributed?
- How does price compare with the number of beds?
- What relationships exist between important numerical variables?

The analysis started with a dataset containing **20,770 Airbnb listings and 22 variables**, including location, room type, price, minimum nights, reviews, availability, rating, bedrooms, beds and baths.

The first challenge was data quality. The dataset contained missing values, duplicate records and data-type issues. Missing rows were removed, duplicates were eliminated, and identifiers such as `id` and `host_id` were converted to appropriate object types.

After cleaning, I explored the distribution of prices and availability and specifically filtered listings with prices below **\$1,500** for the price-focused visual analysis to reduce the influence of extreme values.

One of the most interesting findings was the difference in average pricing across NYC neighbourhood groups:

Neighbourhood Group	Average Price
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Manhattan	\$204.15
Brooklyn	\$155.14
Queens	\$121.68
Staten Island	\$118.78
Bronx	\$107.99

The project also introduced a new **price per bed** feature to understand the cost relative to accommodation capacity. Manhattan had the highest average price per bed at **\$138.71**, followed by Brooklyn at **\$99.79**, Queens at **\$76.34**, Bronx at **\$74.71**, and Staten Island at **\$67.73**.

The analysis then moved beyond simple averages. Bar plots were used to examine price by neighbourhood and room type, scatter plots explored the relationship between reviews and price, pair plots examined multiple numerical relationships, and geographical scatter plots visualized listing locations using latitude and longitude.

Finally, a correlation heatmap was created using numerical variables such as price, minimum nights, number of reviews, reviews per month, availability and beds.